Milestone 2 evidence: glasses feasibility results

RayNeo X3 Pro glasses with our own Kotlin apps; live measurements, no replay or simulation. Laptop on the same home network (Ethernet to the router during the streaming runs). Glasses worn during the IMU runs. Code: https://github.com/hyunkyumkim/seeing-together/tree/ba91f18b6ed00e949fc96c284a15a3e857bcfeab

1. IMU streaming over Wi-Fi (UDP, 100 Hz requested)

20 s runs (sample loss checked by sequence number):

Run	**Sensor**	**Samples received**	**Lost**	**Generated rate**	**Device timestamp spacing**
20 s (imu-test/test.csv)	Accelerometer	2,193	0 (0.0%)	109.5 Hz	9.13 ms (mean = max)
20 s (imu-test/test.csv)	Gyroscope	2,193	0 (0.0%)	109.5 Hz	9.13 ms (mean = max)
20 s repeat after crash fix	Accel / Gyro	2,198 / 2,199	0 (0.0%)	109.5 Hz	9.13 ms

Wi-Fi drop run (90 s):

Item	**Result**
Procedure	Glasses Wi-Fi turned off about 19 s into the run for ~17 s (estimated), then turned back on (adb svc wifi)
App survived	Yes: same process ID before and after; on-screen sendErr = 4703 caught send failures, queueFull = 0
Recovery	Streaming resumed by itself, no restart
Data gap	39.2 s (19.4 s to 58.6 s; includes Wi-Fi reconnect time)
Loss	4,292 of 9,866 accelerometer samples (43.5%), all inside the gap; 0 lost outside it (no buffering)

2. AprilTag detection from glasses camera stills

Camera ID 0 (back-facing), 1920x1080 JPEG via Camera2. tag36h11 ID 0, black square ~20 cm, shown full screen on a monitor. OpenCV 5.0.0 aruco detector, default parameters. Distances and angles estimated by eye.

Distance	Viewing angle	Stills	Detected (ID 0)	False IDs
0.5 m	head-on	3	3 / 3	0
1 m	head-on	3	3 / 3	0
2 m	head-on	3	3 / 3	0
3 m	head-on	3	3 / 3	0
1 m	~30°	3	3 / 3	0
1 m	~60°	3	3 / 3	0
Total		18	18 / 18	0

3. Camera frame streaming latency

Camera2 YUV frames, JPEG-encoded on the glasses and sent over TCP to the laptop, 40 s per run, glasses pointed at the clock window. Latency is measured with an on-screen AprilTag clock (ID changes every 50 ms) and covers camera exposure, JPEG encoding, transfer and the monitor's display delay. Frames not delivered are counted from sequence-number gaps.

Link	Size	Delivered fps	Not delivered	Latency median / p95	Samples (n)
Wi-Fi	640x480	29.8	0.8%	106 / 133 ms	590
Wi-Fi (repeat)	640x480	29.8	0.7%	100 / 122 ms	563
Wi-Fi	1280x720	18.6	38.0%	135 / 168 ms	403
Wi-Fi	1920x1080	9.5	68.3%	206 / 244 ms	178
USB	640x480	29.5	1.5%	96 / 118 ms	540
USB	1280x720	18.5	38.4%	127 / 159 ms	382
USB	1920x1080	9.7	67.6%	183 / 212 ms	127

Bottleneck check (glasses on-screen counters, read by eye during the runs): YUV-to-NV21 conversion took 4–6 / 13 / 20–36 ms and JPEG encoding 4–6 / 18 / 60–69 ms per frame at 640x480 / 1280x720 / 1920x1080. Lost frames were dropped because the encoder was still busy; over USB the send queue never overflowed (0 queue drops at every size). Over Wi-Fi at 1280x720 some frames were also dropped at the send queue; with JPEG quality 50 (45 KB

instead of 67 KB per frame) queue drops fell to 0 but the rate stayed at 19.6 fps, so the network was not the limit.

4. AprilTag detection on the stream

Tags 0, 1 and 2 (tag36h11, black square about 15 cm) side by side on a monitor; glasses held by hand at the given distance, Wi-Fi, 30 s per run. Detection rate = frames in which the tag was found, out of the frames checked by the receiver (it checks the newest frame only). Each detection was also saved as a timestamped observation for the shared memory.

Run	**Distance**	**Size**	**Delivered fps**	**Tag 0**	**Tag 1**	**Tag 2**
tag_05m	0.5 m	640x480	30.0	100%	100%	100%
tag_1m	1 m	640x480	29.9	99.8%	100%	100%
tag_1m_b	1 m	640x480	30.0	99.8%	99.1%	100%
tag_2m	2 m	640x480	30.0	41.4%	40.5%	85.2%
tag_2m_b	2 m	640x480	29.9	21.8%	17.7%	58.1%
tag_2m_c	2 m	640x480	30.0	56.4%	64.1%	91.8%
tag_2m_720	2 m	1280x720	26.8	62.6%	50.6%	83.0%
tag_2m_720_b	2 m	1280x720	27.1	86.7%	68.6%	93.3%

5. Limitations

Short runs on home Wi-Fi only. The camera opens only when the app is in the foreground with the screen on (otherwise CAMERA_DISABLED). Tags on a screen, not printed; detection only, no pose accuracy; single tag; the 3 stills per condition are near-duplicates, not independent trials. The detection limit beyond 3 m / 60° has not been found yet. Frame streaming: latency has about 50 ms resolution and includes the monitor's display delay; one scene on home Wi-Fi; only 640x480 over Wi-Fi was repeated; the 1080p runs have fewer latency samples; audio was not tested. Early runs with a stale app connection and a static ghost tag were discarded and re-measured. An earlier build copied the color planes pixel by pixel and reached only about 14 fps at 640x480; all results above are from the fixed build (row-wise copy, conversion 4–6 ms). Tag detection on the stream: hand-held, distances by eye, one to three runs per condition; at 2 m the rate varied widely between runs at the same size, so these runs cannot tell whether 1280x720 is better than 640x480.